

Scatterplots of the Mordell–Weil generators for high rank elliptic curves

Michael Rubinstein
University of Waterloo

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Abstract

We plot the Mordell–Weil generators of a few million elliptic curves over $\mathbf{Q}$ of rank 5 to 13, normalized by the curve’s own c_4 , and find that the resulting scatterplots are governed by a single parameter which we identify with the j -invariant. Every visible band is a level set of that parameter, and the sharpest feature is the nodal cubic $\eta^2 = (\xi - 3)^2(\xi + 6)$, the discriminant-zero member of the normalized family, on which curves of low conductor accumulate. We explain why that accumulation is forced rather than accidental, and separate the features of the plots that are arithmetic from those that are artifacts of how the curves were found.

1 Introduction

These notes record scatterplots of the Mordell–Weil generators of many elliptic curves over $\mathbf{Q}$ of rank between 5 and 13, and explain the features those plots show. Figure 2, on the last page, is the object of interest: one panel per rank, drawn from the 50,000 curves, or fewer, of least conductor with $c_4 > 0$ at each rank, which is 344,042 curves and 2,740,006 generators in all, selected from a database of 2,752,142 curves of these ranks. Every curve in it has proven rank, with independent generators exhibited and checked on the curve.

The plots are made by scaling each generator by its own curve’s c_4 . Writing the curve in the form (3) and normalizing as in (4), each generator becomes a point (ξ, η) , and the resulting cloud is not uniform. The points accumulate along the single curve

$$\eta^2 = \xi^3 - 27\xi + 54 = (\xi - 3)^2(\xi + 6), \tag{1}$$

a cubic with a node at $(3, 0)$ and a branch point at $(-6, 0)$. It is worth being precise that (1) is not itself an elliptic curve: it is the discriminant-zero member of the family $\eta^2 = \xi^3 - 27\xi + \tau$ that the normalization produces, so it is singular, and that degeneracy is exactly why the generators pile up on it. We call it the governing curve.

The explanation occupies the rest of these notes. Sections 2 and 3 set up the normalization and identify the single parameter τ on which the whole picture depends, showing it to be a function of the j -invariant alone. Section 4 explains the accumulation: small $\varepsilon = 54 - \tau$ forces the discriminant to be small relative to the height, so a search for curves of low conductor is, without intending to be, a search near (1). Section 5 separates the features that are arithmetic from those that are artifacts of how the curves were found, and §6 asks how much of the picture depends on which basis of generators is returned.

This is an ongoing computation. The searches are being extended to higher ranks, and the method used to find the curves will be described elsewhere. The database behind Figure 2 is a snapshot of it.

The algorithm for finding the curves is the author's; Claude implemented it, and ran it on roughly a thousand cores of the University of Waterloo MFCF cluster. Claude worked as a collaborator in analyzing and explaining the scatterplots, and in assisting to prepare these notes.

2 Normalization

Let $E/\mathbf{Q}$ have invariants c_4, c_6 and minimal model $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. Put

$$X = 36x + 3b_2, \quad Y = 108(2y + a_1x + a_3), \quad b_2 = a_1^2 + 4a_2, \quad (2)$$

so that (X, Y) satisfies

$$Y^2 = X^3 - 27c_4X - 54c_6. \quad (3)$$

Scale by the curve's own c_4 :

$$\xi = \frac{X}{\sqrt{|c_4|}}, \quad \eta = \frac{|Y|}{|c_4|^{3/4}}. \quad (4)$$

Both are invariant under $(c_4, c_6, x, y) \mapsto (u^4c_4, u^6c_6, u^2x, u^3y)$, so every Weierstrass model of E gives the same point of the (ξ, η) plane. For $c_4 > 0$, (3) becomes

$$\eta^2 = \xi^3 - 27\xi + \tau, \quad \tau = \frac{-54c_6}{|c_4|^{3/2}}, \quad (5)$$

and for $c_4 < 0$ the sign of the linear term reverses,

$$\eta^2 = \xi^3 + 27\xi + \tau. \quad (6)$$

After the normalization (4) the entire family therefore depends on the single real parameter τ .

3 τ is the j -invariant

The parameter τ of (5) is not a new invariant. Squaring its definition,

$$\tau^2 = \frac{2916 c_6^2}{|c_4|^3}, \quad (7)$$

and the discriminant identity $c_4^3 - c_6^2 = 1728\Delta$ lets us remove c_6 ,

$$c_6^2 = c_4^3 - 1728\Delta. \quad (8)$$

Substituting (8) into (7) gives

$$\tau^2 = 2916 \frac{c_4^3 - 1728\Delta}{|c_4|^3} = 2916 \frac{c_4^3}{|c_4|^3} \left(1 - \frac{1728\Delta}{c_4^3}\right). \quad (9)$$

Since $|c_4|^3 = |c_4^3|$, the prefactor $c_4^3/|c_4|^3$ is $\text{sgn}(c_4)$, and by $j = c_4^3/\Delta$ the bracket is $1 - 1728/j$. Hence

$$\tau^2 = 2916 \text{sgn}(c_4) \left(1 - \frac{1728}{j}\right), \quad (10)$$

which for $c_4 > 0$, the case of (5), is

$$\tau^2 = 2916 \left(1 - \frac{1728}{j} \right). \quad (11)$$

The sign in (10) is not a blemish but a consistency check: for $c_4 < 0$ the left side is nonnegative, so $1728/j \geq 1$, and indeed $c_4^3 < 0$ with $j = c_4^3/\Delta$ forces $0 < j \leq 1728$ exactly when $\Delta < 0$. The two cases (5) and (6) therefore occupy disjoint ranges of j .

Thus τ carries exactly the information of the j -invariant, and each level set of τ in the (ξ, η) plane is the locus of a single j . Quadratic twists have the same τ , so the normalization (4) quotients out twisting: all twists of E have their generators on one and the same curve in the picture, at different positions along it.

Two values of τ are distinguished. From (11), $\tau = \pm 54$ corresponds to $j = \infty$, that is to $\Delta = 0$, and there the cubic (5) degenerates:

$$\tau = +54 : \quad \eta^2 = \xi^3 - 27\xi + 54 = (\xi - 3)^2(\xi + 6), \quad (12)$$

$$\tau = -54 : \quad \eta^2 = \xi^3 - 27\xi - 54 = (\xi + 3)^2(\xi - 6). \quad (13)$$

These are nodal cubics, with nodes at $\xi = 3$ and $\xi = -3$ respectively. They are rational curves, parametrized by

$$\xi = s^2 - 6, \quad \eta = s^3 - 9s \quad (14)$$

in the case (12). Figure 1 plots the two of them. We refer to them as the governing curves.

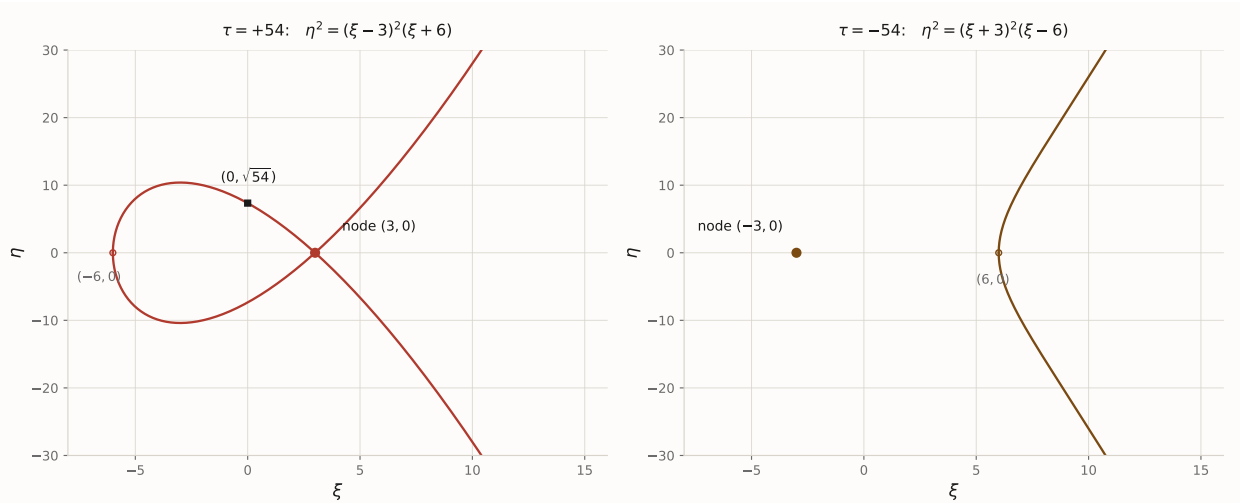


Figure 1: The two governing curves. Left: $\tau = +54$, node at $(3, 0)$, crossing $\xi = 0$ at $\eta = \sqrt{54} = 7.348$. Right: $\tau = -54$, node at $(-3, 0)$. Both are the $\Delta = 0$ degenerations of (5).

4 Why the governing curves are relevant to generators

A generator P of $E(\mathbf{Q})$ is a rational point of E , so its image under (2) and (4) lies on the curve (5) belonging to that E . Nothing else is possible: every point of the scatterplot lies on the level set of its own τ . Consequently the plot is a superposition of level sets, one per curve, weighted by how

many generators each contributes and by where along the level set they fall. A band of high or low density can only be a level set along which the weight is anomalous.

The governing curves are where the weight is most anomalous, for the following reason. Write $\varepsilon = 54 - \tau$. Substituting $j = c_4^3/\Delta$ into (11) gives $\tau^2 = 2916(1 - 1728\Delta/c_4^3)$, and solving for Δ gives $\Delta = c_4^3(2916 - \tau^2)/(2916 \cdot 1728)$. Since $2916 - \tau^2 = \varepsilon(108 - \varepsilon)$, this is exactly $\Delta = c_4^3\varepsilon(108 - \varepsilon)/5038848$, and discarding ε^2 against 108ε leaves

$$\Delta = \frac{c_4^3 - c_6^2}{1728} \approx \frac{c_4^3 \varepsilon}{46656} \quad (\varepsilon \rightarrow 0), \quad (15)$$

so ε small correlates with $|\Delta|$ small relative to the height. Since the conductor divides the discriminant, a curve of small conductor at a given height is a curve with ε small, hence one whose level set is close to the first governing curve (12). Searching for low conductor is therefore searching near the first governing curve, and the accumulation is not a coincidence of the search but a restatement of what the search achieves.

Inverting the normalization once, for later use: since (5) holds at every plotted point, τ is a function of position in the plane,

$$\tau(\xi, \eta) = \eta^2 - \xi^3 + 27\xi. \quad (16)$$

Each level set of τ is therefore a single curve in the (ξ, η) plane, and reading τ off a point is a matter of evaluating (16).

5 Which features are arithmetic and which are artifacts

The dark cubic. The level set $\tau = 54$, equation (12). It appears as an enhanced band at every rank, because every curve of the search is selected for low conductor and therefore for small ε .

A concentration on the vertical axis near $\eta \approx 7$. On the line $\xi = 0$ equation (16) collapses to $\eta^2 = \tau$, so along that line $\eta = \sqrt{\tau}$ identically. The distribution of η on the vertical axis is therefore the distribution of τ carried through a square root, and nothing else. Since τ piles up at 54, η piles up at $\sqrt{54} = 7.348$. The arithmetic is checked by the numbers: on the strip $|\xi| < 0.5$ at ranks 8 to 10, over the 93,732 plotted points it contains, the median η is 7.2366, corresponding to $\tau = 52.37$, and 24.7% of the strip lies between $\sqrt{49} = 7.000$ and $\sqrt{59} = 7.681$, against the 30.0% of the 1,299,469 curves of those ranks with $c_4 > 0$ that have $|\tau - 54| < 5$.

A hollow oval between $\xi = -5.2$ and $\xi = 0$. This is the most conspicuous empty region in the plots, and it has an exact boundary. Inverting (16), a plotted point has $\tau < 0$ precisely when $\eta^2 < \xi^3 - 27\xi$, and $\xi^3 - 27\xi = \xi(\xi^2 - 27)$ is positive exactly on $(-\sqrt{27}, 0) \cup (\sqrt{27}, \infty)$. On the first of those intervals the region is bounded, an oval running from $\xi = -\sqrt{27} = -5.196$ to $\xi = 0$ whose apex is at $\xi = -3$, where $\eta = \sqrt{54} = 7.348$. Those two numbers are the extent of the hole.

It is empty because $\tau = -54c_6/c_4^{3/2}$, so for $c_4 > 0$ the condition $\tau < 0$ is exactly $c_6 > 0$, and the curves plotted here have $c_6 < 0$. The interior of the oval is therefore not a region the data declines to populate, it is a region the data cannot reach. The same boundary bounds a second empty region under the right-hand branch, beyond $\xi = \sqrt{27}$, for the same reason. Both are artifacts of the sign restriction.

6 Choice of generators

A basis of $E(\mathbf{Q})/E(\mathbf{Q})_{\text{tors}}$ is not canonical, so it is fair to ask how much of the picture depends on which basis is returned. The answer is that the support does not depend on it at all and the

distribution along the support does. Every point of $E(\mathbf{Q})$ lies on the same level set, so no choice of basis can move a point off the curve (5). Replacing a basis by another changes only where along that curve the points fall. For the rank 6 record curve, the basis returned by `ellrank` has

$$\xi \in \{-5.89, -4.05, -2.21, -0.06, 15.58, 21.11\}, \quad (17)$$

while an LLL-reduced basis for the height pairing has

$$\xi \in \{-5.99, -4.89, -3.23, 4.98, 6.28, 33.68\}, \quad (18)$$

and the doubles $2P_i$ reach $\xi = 194.97$. Heights grow under addition, so a basis chosen to minimize height concentrates the points near the node, and any other choice pushes them out along the arms. The plots in this note use the basis returned by `ellrank`, which is height-reduced, and they should be read as showing the smallest points on each curve.

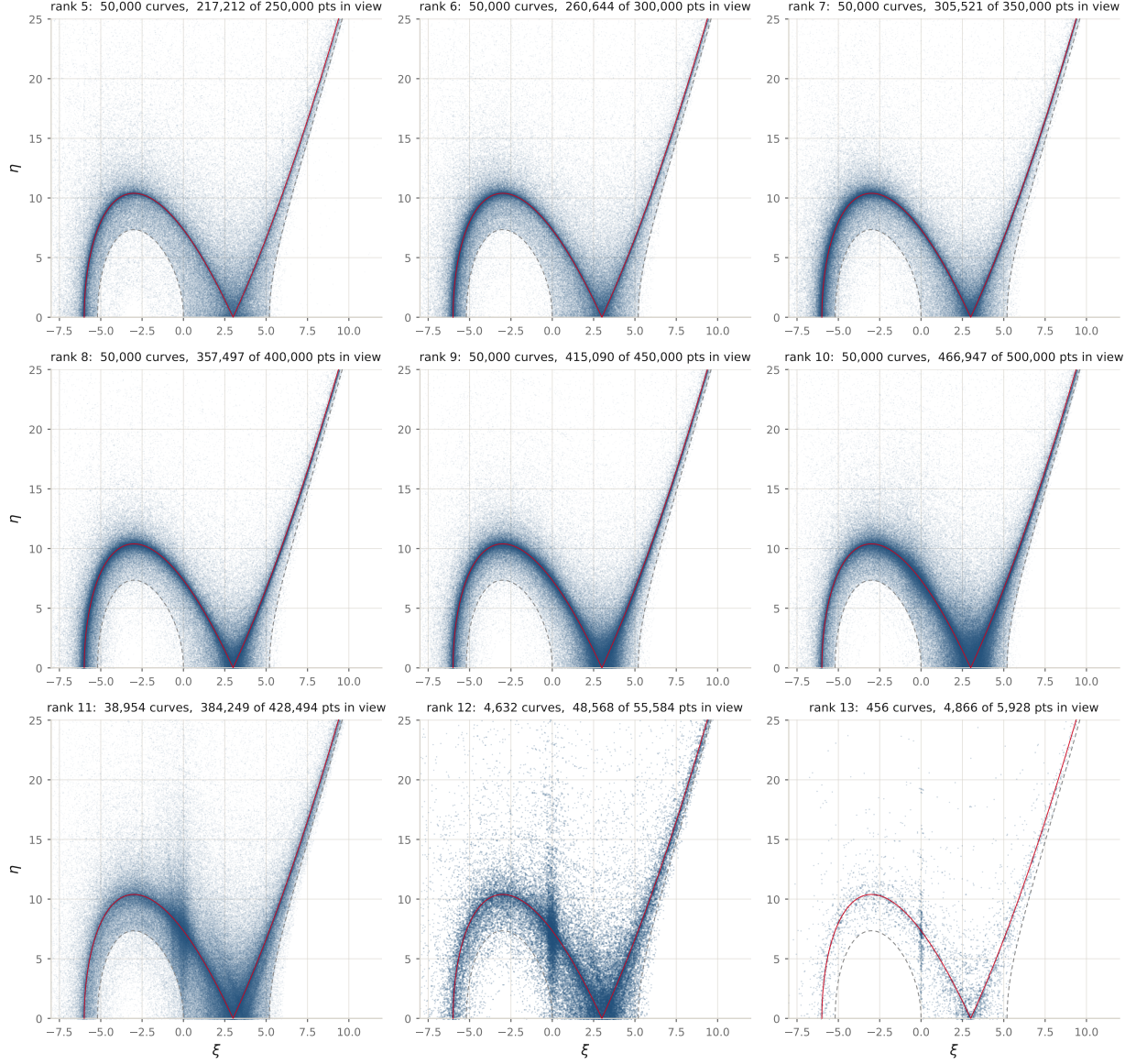


Figure 2: Normalized generators for ranks 5 to 13, from the 50,000 curves of least conductor *among those with $c_4 > 0$* at each rank, or from all of them at the three highest ranks, where the data holds fewer: 38,954 at rank 11, 4,632 at rank 12 and 456 at rank 13. Each panel is titled with the number of curves, the number of generators drawn, and the number available after the restriction to $c_4 > 0$. The two do not agree because the axes are cut at $-8 \leq \xi \leq 12$ and $0 \leq \eta \leq 25$, and the omitted generators run far out along the arms of the governing curve, where $\eta \sim |\xi|^{3/2}$: their median ξ is 15.1 and they reach $\xi = 1.7 \cdot 10^7$ and $\eta = 7.2 \cdot 10^{10}$. Plotting them would compress the structure near the node to a point. The thin red line is the governing curve $\tau = +54$, equation (12). The grey dashed curve is the locus $\tau = 0$, that is $\eta^2 = \xi^3 - 27\xi$; it bounds the hollow oval on the left and a second nearly empty region under the right-hand branch, both being the image of the condition $c_6 < 0$. “Nearly” because a small fraction of the curves plotted here have $c_6 > 0$.